

B.Tech IV Year I Semester (R20) Regular & Supplementary Examinations December/January 2024/2025

HEALTH, SAFETY & ENVIRONMENTAL MANAGEMENT

(Common to EEE, ME, ECE, CSE, IT, FT, AI&DS, CSE(AI), CSE(DS), CSE(AI&ML), CSE(IoT), CSE(CS), CS&D and AI&ML)

Time: 3 hours

Max. Marks: 70

PART – A
(Compulsory Question)

- 1 Answer the following: (10 X 02 = 20 Marks)
- | | |
|--|----|
| (a) State the importance of safety regulations. | 2M |
| (b) Explain the role of safety inspector in inspection. | 2M |
| (c) Differentiate Pollution and Pollutant. | 2M |
| (d) Identify the steps involved in preliminary hazard analysis. | 2M |
| (e) Describe Functional Resonance Accident Model (FRAM). | 2M |
| (f) List the questions which helps in accident investigation. | 2M |
| (g) Describe Critical incidence technique of accident analysis. | 2M |
| (h) Write basic components of an accident investigation process. | 2M |
| (i) Compare voluntary and involuntary risk. | 2M |
| (j) Define risk estimation. | 2M |

PART – B
(Answer all the questions: 05 X 10 = 50 Marks)

- 2 Describe in detail about general safety measures to be adopted. 10M
- OR**
- 3 (a) Describe various groups or individuals who might be part of a disaster control organization. 5M
(b) Explain the factors for consideration of safety warning systems. 5M
- 4 (a) Explain in detail about hazard monitoring. 5M
(b) Explain in detail about toxic gases that cause air pollution. 5M
- OR**
- 5 (a) Discuss in detail hazard control methods. 5M
(b) Describe the ways by which you can protect the environment from toxic components of chemical industry. 5M
- 6 (a) Write down the different types of analysis of accident. 5M
(b) Evaluate the root cause behind occurrence of fire accidents. 5M
- OR**
- 7 Discuss in detail toxic release and dispersion modelling. 10M
- 8 (a) Discuss the term HAZOP. Explain in detail how is HAZOP conducted. 5M
(b) What are the different parameters of safety in chemical industry? 5M
- OR**
- 9 Write a detailed note on: 10M
(i) Preliminary Hazard Analysis (PHA),
(ii) Safety measures in design and process operations.
- 10 (a) Discuss steps involved in risk estimation. 5M
(b) Explain the approaches for establishing acceptance level. 5M
- OR**
- 11 Elaborate the sources of risk and explain the various methods of risk measurement. 10M

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Time: 3 hours

Max. Marks: 70

PART – A
(Compulsory Question)

- 1 Answer the following: (10 X 02 = 20 Marks)
- | | |
|--|----|
| (a) Name the basic terms used in safety and environmental management practices. | 2M |
| (b) What do you understand by Safety health and environmental management? | 2M |
| (c) Define Pollution. | 2M |
| (d) What is the meaning of alteration in physical and chemical characteristics of water? | 2M |
| (e) What is Modeling? | 2M |
| (f) Name the effects of toxic release from the human activity. | 2M |
| (g) What is the full form of PHA? | 2M |
| (h) What is Sprinkler System? | 2M |
| (i) What is Fatality risk assessment? | 2M |
| (j) Define Risk assessment. | 2M |

PART – B

(Answer all the questions: 05 X 10 = 50 Marks)

- | | | |
|-----------|---|----|
| 2 | (a) Discuss in brief safety for organizations. | 5M |
| | (b) Write a note on importance of Environmental management. | 5M |
| OR | | |
| 3 | (a) Explain in brief safety assurance and assessment. | 5M |
| | (b) Explain in brief safety in design and operation. | 5M |
| 4 | (a) Define Hazard and discuss hazard classification. | 5M |
| | (b) Write a note on Hazard evaluation and hazard control. | 5M |
| OR | | |
| 5 | (a) Explain causes for Water pollution. | 5M |
| | (b) Write a note on Environmental Monitoring. | 5M |
| 6 | (a) Explain in brief about Accidents modeling. | 5M |
| | (b) Write a note on Release modeling. | 5M |
| OR | | |
| 7 | (a) Explain the importance of fire and explosion modeling. | 5M |
| | (b) Discuss in brief dispersion modeling. | 5M |
| 8 | (a) Write a note on Accident investigations. | 5M |
| | (b) Explain in brief the concept of HAZOP. | 5M |
| OR | | |
| 9 | (a) Write a note on safety measures in design and process operations. | 5M |
| | (b) Explain what are the steps in fire prevention. | 5M |
| 10 | (a) Write a note on Hazard assessment. | 5M |
| | (b) Explain the brief quantified risk assessment. | 5M |
| OR | | |
| 11 | (a) Write a note on Risk management principles. | 5M |
| | (b) Explain different methods of Risk management. | 5M |
